Casey Barton

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PROFILE

I come at this role from both ends of the construction workflow. My working background is home inspection — permits, site walkthroughs, materials, punch lists, and photo documentation on real jobsites — so I have operated the document and jobsite workflows Fieldwire digitizes. My engineering practice is applied AI systems: evaluation harnesses, failure-mode recovery, and trace intelligence across agent and retrieval pipelines, with **65** authored MCP connectors and typed contracts governing every tool boundary.

Three years as a founder running an evidence-grade agent control plane from Honolulu: adapting and evaluating modern models (VLMs, LLMs, RAG architectures, agents) against production tasks rather than training from scratch; golden datasets and behavioral assertions as the quality discipline; CI regression gates that fail closed; per-task latency and token-cost instrumentation that drives optimization. Python primary, strong APIs, production code discipline.

CORE COMPETENCIES

Construction domain: home-inspection background — permits · site walkthroughs · materials assessment · punch lists · photo / site documentation · plan reading · jobsite coordination workflows

AI & Evaluation: VLM / LLM / RAG / agent pipelines · model adaptation & evaluation for production tasks · evaluation harnesses · LLM-as-a-judge with written rubrics · golden datasets · behavioral assertions · quality scoring · trace intelligence · data curation & quality assessment · failure-mode instrumentation · MCP servers & connectors · structured outputs · guardrails · fail-closed recovery

Production: Python (primary) · TypeScript · SQL · Bash · REST APIs · JSON Schema / Pydantic-style validation · GitHub Actions (CI/CD) · Postgres · high-performance pipeline optimization · token budgets · retry logic · observability & run logging · latency/cost benchmarking · cloud deployment patterns · documentation engineers actually read

EXPERIENCE

Founder / Applied AI Systems Architect 2023 – Present

GlacierEQ — Honolulu, HI

- Adapt and evaluate modern models for production tasks — VLM, LLM, RAG, and agent pipelines judged on real site-like task criteria: adapt existing models and architectures to task demands, own components from conception to production deployment.
- Lead data curation and data quality assessment as a standing discipline: golden datasets, behavioral assertions, and judge criteria with written rubrics; quality gates that fail closed before model updates ship.
- Build and optimize high-performance ML/agent pipelines: per-task latency and token-cost instrumentation on every run; model selection per task complexity; trace intelligence naming the slowest calls, redundant tool use, and expensive segments — findings fed back into configuration.

- Design multimodal understanding workflows: image plus document inputs routed through model pipelines with structured outputs and strict schema validation on every tool boundary — **65** MCP connectors exposing governed context with evidence boundaries and audit receipts.
- Design agent systems for failure: instrument the failure modes found in production — stuck loops, bad retrievals, tool misuse, context blowouts — and build the catchers: replay loops, fail-closed gates, behavioral assertions, recovery restricted to evidence declared before reassessment.
- Run an evidence-grade agent control plane across a **1,413-repository** estate: typed handoff contracts, guardrails, recoverable runs — backed by a **989-test** suite, **95/95** verification checks green.

EDUCATION

University of Hawaii at Manoa 2016
Bachelor of Science

AWS Cloud Institute 2025
Cloud Computing program

Colorado State University
Undergraduate coursework

SELECTED PROOF

Job-App Helix — evaluation-gated campaign engine

- Attributable campaign engine with immutable evidence models and fail-closed **GO / NO-GO** decisions; the eval gate decides, not the agent.
- Judge criteria with written rubrics behind every automated quality decision — the quality-scoring discipline Fieldwire's AI role centers on.

Construction-domain edge

- Home-inspection background: permits, site walkthroughs, materials assessment, punch lists, photo and site documentation on real jobsites — firsthand knowledge of the workflows Fieldwire's AI products serve.
- Models evaluated against task-grounded criteria, not benchmark proxies — the same discipline jobsite AI demands.

SKILLS

Languages: Python · TypeScript · SQL · Bash · JavaScript / Node.js

AI / Evaluation: VLM / LLM / RAG / agent pipelines · model adaptation & evaluation · eval harnesses · quality scoring · LLM-as-a-judge · golden datasets · trace intelligence · MCP servers & connectors · function calling · structured outputs · guardrails · fail-closed recovery

Production: conception-to-production ownership · data curation & quality assessment · JSON Schema / Pydantic-style validation · HTTP APIs · CI regression gates · observability & run logging · latency/cost tradeoff analysis · cross-layer debugging · documentation & handover

Evidence boundary: portfolio repositories are independent work. VLM/multimodal depth is estate-verified (adaptation + evaluation pipelines) — no training-from-scratch or research-lab claims. No Fieldwire-internal production claims. GitHub is the forge of record.